

EngageIQ — Smart Engagement Opportunity Scorer

Shivneel Prasad · BAX-423 Big Data · Spring 2026

Live URL: <https://engageiq-bax423-final.streamlit.app/>
GitHub: <https://github.com/shprasa/engageiq-bax423-final>

1. System architecture & pipeline

Pipeline: GitHub API + Hacker News API → URL dedup on ingest → DuckDB store → TF-IDF/SVD embeddings → cosine ANN retrieval → multi-stage ranker (relevance, health, visibility, effort, recency) → reinforcement learning (Thompson-sampling contextual bandit) from engage/skip/bookmark feedback → Streamlit dashboard with trend analytics and CSV/PDF brief export. Offline snapshot: 10000 rows (1964 live API + 8036 synthetic backup), 15 domains. Live sources: {'hackernews': 1036, 'github': 928}.

2. BAX-423 techniques (2 required)

1. Recommendation system: TF-IDF/SVD embeddings + NearestNeighbors retrieval; multi-stage ranking with NDCG@10 per persona. **2. Reinforcement learning:** contextual multi-armed bandit (Thompson sampling) — state=user profile+history, actions=domain recommendations, rewards from engage/bookmark/skip. 60-round benchmark: NDCG@10 with RL 1.000 vs without 1.000; cumulative reward 60 vs 15; reward $\Delta(\text{last } 10)=+0.700$. Ingest: {'hackernews': 5036, 'github': 4964}; 15/15 domains.

3. Persona results (top-10 checks)

Persona	NDCG@10	Checks	Overall
Sofia	0.933	GFI=7, C++/Rust=0	PASS
David	0.000	Infra=10/10	PASS
Lina	0.000	Visibility $\geq$ relevance	PASS
Raj	1.000	DevTools=10/10	PASS

4. Persona × capability pass/fail (required)

Persona	Multi-source ingest	Embeddings + ANN	Scoring + multi-stage	Adaptive learning / RL	Batch analytics + trend	Dashboard + brief expo
Sofia	PASS	PASS	PASS	PASS	PASS	PASS
David	PASS	PASS	PASS	PASS	PASS	PASS
Lina	PASS	PASS	PASS	PASS	PASS	PASS
Raj	PASS	PASS	PASS	PASS	PASS	PASS

5. Limitations

Synthetic backup rows (GitHub + HN only) ensure 10k offline grading. Embeddings use TF-IDF/SVD vs Sentence-BERT/FAISS for deploy reliability. Suggested actions are template-based.